Claudio Camolese

Curriculum Vitae

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EDUCATION

École Normale Supérieure Paris-Saclay, France
2026-2027 - Incoming student
Master of research in Applied Mathematics, MVA track - Mathematics, Vision and Learning

Politecnico di Torino, Italy
École Nationale Supérieure d'Informatique et de Mathématiques Appliquées, France
2024-2026
Double Master's degree program, specializing in Artificial Intelligence.
Overall GPA: 3.6/4.0, first class honours
Relevant courses: *Robotics*, *Robot learning*, *Computer Vision*, *Deep Natural Language Processing*, *GPU computing*, *Generative multimodal AI*, *Machine learning and Deep learning*, *Mathematics in machine learning*

Politecnico di Torino, Italy
2021-2024
Bachelor's degree in Physical Engineering.
Overall GPA: 3.3/4.0
Relevant courses: *Advanced Methods for Physics*, *Quantum physics*, *Solid state physics*, *Advanced electromagnetics*

PUBLICATIONS

Camolese, C., Averta, G., Wysocki, O., Wang, G., PhiSVLA: Physics-Informed Safety for Vision Language Action manipulation. IEEE/CVF Conference on Computer Vision and Pattern Recognition (manuscript in preparation).

AI & ROBOTICS EXPERIENCE

University of Cambridge and TUM — *October 2026 – February 2027*
Research Intern – Supervisors: Prof. Wysocki, Prof. Wang

Incoming Research Intern on *Interactive World-Action Models for Open-World Robotic Manipulation.*

University of Cambridge, CV4DT team — *February 2026 – Present*
Research Intern – Supervisors: Prof. Wysocki, Prof. Wang, Prof. Averta, Prof. Pineda

Physics-Informed Safety: Developed a plug-and-play safety layer for Vision-Language-Action models that applies minimal corrections to unsafe actions while preserving the intended task.

Vision-Language Perception: Integrated Qwen2.5-VL and SAM3 to identify task-relevant objects and exclude manipulation targets and placement goals from the obstacle representation.

SafeLIBERO Evaluation and Metric Design: Evaluated the framework on safety-critical manipulation tasks involving multiple obstacles and constrained passages, and proposed a new metric for measuring collision-free task success. Relative to the frozen $\pi_{0.5}$ policy, PhySVLA improves the Collision Avoidance Rate (CAR) by 325.2% and the Safe Success Rate (SSR) by 266.0%. Compared to the current state-of-the-art execution-time safety approach using the same 0.5 backbone, PhySVLA further improves CAR by 10.2% and SSR by 15.1%.

Politecnico di Torino, Squadra Corse Driverless — *September 2024 – October 2025*
Team member - Perception and AI engineer

LiDAR-Based State Estimation: LiDAR-Based State Estimation: Developed components of the vehicle state-estimation pipeline with an Ouster OS0 sensor, leveraging ROS2, Eigen, Sophus, oneTBB, and Open3D.

Reinforcement Learning Integration: Integrated a reinforcement-learning agent into MATLAB and Simulink for operation on pre-mapped tracks, enabling the vehicle to exploit known track geometry and reducing its reliance on online perception.

Formula SAE Driverless: Supported the development and testing of the autonomous racing system and competed in Formula SAE Driverless competitions.

Politecnico di Torino, UROP Research Programme *February 2025 – October 2025*
Research Student – Supervisors: Prof. Daniele Pagliari, Dr. Luca Benfenati

Memory-Efficient LLM Inference: Investigated inference strategies for deploying reasoning Large Language Models on memory-constrained edge devices using DeepSeek-R1.

Recursive Reasoning Pipeline: Designed a recursive inference pipeline that iteratively feeds intermediate outputs back into the model, decomposing the reasoning process into sequential steps that remain within the device memory constraints.

AI & ROBOTICS PROJECTS

RGB-D 6D Object Pose Estimation: Developed an end-to-end pipeline for estimating object position and orientation from RGB-D observations. Evaluated the approach on the LINEMOD dataset using the ADD(-S) metric, and adapted it for real-time traffic-cone pose estimation in the Squadra Corse Driverless simulator while the vehicle was in motion.

Semantic Scene Understanding for Autonomous Drones: Implemented a DeepLabV3+ architecture for semantic segmentation of aerial imagery, enabling pixel-level scene understanding for autonomous navigation and environmental perception.

Sim-to-Real Reinforcement Learning: Developed robust control policies for the MuJoCo Hopper under source–target dynamics shifts. Applied Uniform Domain Randomization to robot link masses and evaluated its effectiveness in reducing performance degradation when transferring policies across different dynamics.

Generative Models for Image Synthesis: Designed and implemented diffusion models, flow-matching models, GANs, and VAEs from scratch, including their architectures, training objectives, sampling procedures, and optimization pipelines. Compared the different generative paradigms in terms of training stability, sample quality, and inference behaviour.

AWARDS & ACHIEVEMENTS

UROP@PoliTO Open Badge: Awarded by Politecnico di Torino upon completion of a supervised Undergraduate Research Opportunities Programme project on memory efficient LLM inference for edge devices.

ManagerItalia Academic Excellence Scholarships: Recipient of five merit-based scholarships awarded by Manageritalia and the Fondo Mario Negri in recognition of sustained academic achievement.

Stellantis Student Award: Recipient of a Stellantis merit-based student award for academic excellence.

Imprenditori #GenNext: Selected for the entrepreneurship programme organised by UniCredit and the Club degli Investitori, focused on start-up creation, fundraising, and intellectual property.

SKILLS

Programming: Python, C++, CUDA, MATLAB, Java

AI & Robotics Tools: PyTorch, LIBERO, MuJoCo, PyBullet, ROS 2, Gazebo, RViz, OpenCV, LiDAR

LANGUAGES

Italian: Native speaker

English: IELTS C1

French: DELF B1